

Section 2.5 Independence

Intuitively: We say two events A and B are independent, if knowing that A occurred does not change the probability of the occurrence of B .

Definition (Independence of Two Events)

Events A and B are independent if and only if

$$P(A \cap B) = P(A) \cdot P(B) \quad (1)$$

Note: Clearly if two events A and B are independent then,

$$P(B|A) = P(B) \quad (2)$$

or

$$P(A|B) = P(A) \quad (3)$$

To prove independence of two events, we only need to show one of the three above equations is true.

Eg 1. The probability that a randomly selected student at UMD owns a scooter is 0.55, the probability that a student owns a car is 0.30, and the probability that a student owns both is 0.10. Let

$A = \{\text{a student owns a scooter}\}$

$B = \{\text{a student owns a car}\}.$

Are events A and B independent?

$$P(A) = 0.55, \quad P(B) = 0.3, \quad P(A \cap B) = 0.1$$

$$P(A) \cdot P(B) = 0.55 \cdot 0.3 = 0.165 \neq P(A \cap B)$$

Hence A and B are dependent.

Eg 2. Two events C and D in a sample space S are given. Suppose $P(C) = 0.3$, $P(D|C) = 0.2$, $P(C \cup D) = 0.64$.

- (a) What is $P(C \cap D)$?
- (b) Are events C and D independent?

using multiplication rule

$$P(C \cap D) = P(C) P(D|C) = 0.3 \cdot 0.2 = 0.06$$

$$\begin{aligned} P(D) &= P(C \cup D) - P(C) + P(C \cap D) \\ &= 0.64 - 0.3 + 0.06 = 0.4 \end{aligned}$$

$P(D|C) \neq P(D) \Rightarrow D$ and C are dependent.

Eg 3. Events E and F in a sample space S , with $P(E) = \frac{3}{4}$, $P(F) = \frac{1}{3}$, $P(E \cap F) = \frac{1}{4}$, are give.

- (a) What is $P(E|F)$?
- (b) What is $P(F|E)$?
- (c) Are events E and F independent?

$P(E) \cdot P(F) = \frac{1}{3} \cdot \frac{3}{4} = \frac{1}{4} = P(E \cap F) \Rightarrow E \text{ and } F$
are independent.

Since E and F are independent $\Rightarrow$

$$P(F|E) = P(F) = \frac{1}{3} \quad \text{and} \quad P(E|F) = P(E) = \frac{3}{4}$$

Eg 4. Assume events E and F in the sample space S are independent. $P(E) = 0.7$ and $P(F) = 0.4$. Calculate the followings.

$$(a) P(F \cap E) = P(E) \cdot P(F) = 0.28$$

$$(b) P(F|E) = P(F) = 0.4$$

$$(c) P(F|E^c) = \frac{P(F \cap E^c)}{P(E^c)} = \frac{P(F) - \overbrace{P(F \cap E)}^{P(F)P(E)}}{P(E^c)} = \frac{P(F) \overbrace{[1 - P(E)]}^{P(E^c)}}{P(E^c)} \\ = P(F) = 0.4$$

$$(d) P(F^c|E^c) = \frac{P(F^c \cap E^c)}{P(E^c)} = \frac{1 - P(F \cup E)}{P(E^c)} \\ = \frac{1 - (0.7 + 0.4 - 0.28)}{0.3} = \frac{0.18}{0.3} \\ = 0.6 = P(F^c)$$

If two events A and B are independent then:

- A and B^c are also independent.
- A^c and B are also independent.
- A^c and B^c are also independent.

Warning!

Eg 5. Suppose A and B are mutually exclusive events ($A \cap B = \phi$) with $P(A) \neq 0$ and $P(B) \neq 0$. Are A and B independent?

No. Because if A occurs then B doesn't occur. So occurrence of A affects occurrence of B . formally

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = 0 \neq P(A) .$$
 So A and B are dependent.

Definition (Independence of Three Events)

Events A , B and C are independent if and only if all the followings are true

$$P(A \cap B) = P(A) \cdot P(B),$$

$$P(A \cap C) = P(A) \cdot P(C),$$

$$P(B \cap C) = P(B) \cdot P(C),$$

$$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$$

More generally, n events $A_1, \dots, A_n$ are called independent if the product formula holds for any subcollection of these events.

Note: Pairwise independence doesn't imply independence

Eg 6. Consider the experiment of two coin flips.

A = First coin is head.

B = Second coin is head.

C = both tosses have the same result.

Are A, B and C independent events?

$$A = \{HT, HH\}$$

$$B = \{HH, TH\}$$

$$C = \{HH, TT\}$$

$$A \cap B = A \cap C = B \cap C = A \cap B \cap C = \{HH\}$$

$$P(A) = P(B) = P(C) = \frac{1}{2}$$

$$\left. \begin{aligned} P(A) \cdot P(B) &= \frac{1}{4} = P(A \cap B) \\ P(A) \cdot P(C) &= \frac{1}{4} = P(A \cap C) \\ P(B) \cdot P(C) &= \frac{1}{4} = P(B \cap C) \\ P(A) \cdot P(B) \cdot P(C) &= \frac{1}{8} \neq P(A \cap B \cap C) \end{aligned} \right\}$$

$\Rightarrow$ A, B, and C are dependent.

Eg 7. $P(\text{Exactly one head in 100 coin flips})$

We know coin flips are independent, knowing the result of previous coin flip doesn't affect next coin flip. So we can multiply the probabilities.

out of 100 flip coin one is head

If we assume $P(H) = p$ and $P(T) = 1-p$

$$P(\text{Exactly one head in 100 coin flips}) = \binom{100}{1} p (1-p)^{99}$$